Name: ______________________

Worksheet

Features of linear relationships

Task A: The constant rate of change

1) Classify these relationships into **linear** and non-**linear**.

$y = 0.4x$

$y = 4x^2$

$y = 4x$

$y = 2x^4$

$y^4 = x + 4$

$x + y = 4$

Linear	Non-**linear**

$y = x^{0.4}$

$y = x + 4$

$4y = x$

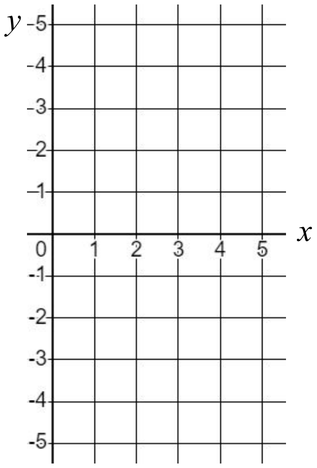
$y = \frac{1}{4}x$

$y = x^4 + 4$

2) Complete the table for the given range of x values. Plot the relationships and show on the graph their constant rate of change.

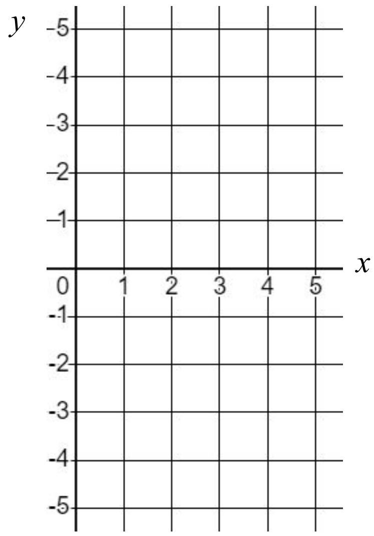
a) $y = 2x - 5$

x	1	2	3	4	5
y					



b) $y = 3 - x$

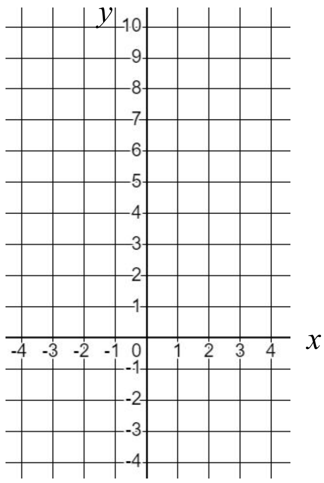
x	1	2	3	4	5
y					



3) This set of coordinates has a linear relationship with a constant rate of change.

Find the missing values, plot the graph, and show the constant rate of change.

x	-4	-2	0	2	4
y	-3				9



Task B: Graphical and algebraic representations

1) Match the coordinate pairs to their respective relationships and graphs.

A $(-4, 1)$ and $(8, 7)$

B $(-4, 5)$ and $(8, -1)$

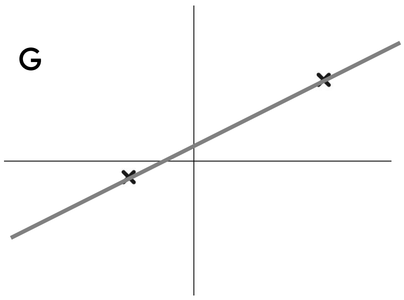
C $(-4, -1)$ and $(8, 5)$

D $y = 3 - \frac{1}{2}x$

E $y = \frac{1}{2}x + 3$

F $y = \frac{x+2}{2}$

G



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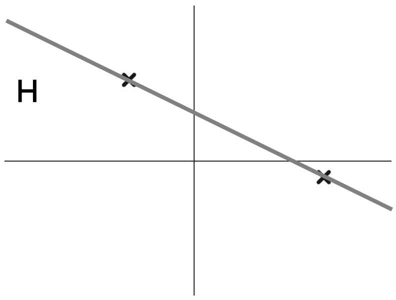

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